



SOUTHEAST UNIVERSITY

Assignment on :

Course Title :

Course Code :

Simple LAN Design and Configuration
with Class A and Class B net ID.
with Inter-network routing.

Submitted by

MD. Kamran Hasan Rafin
Dept. of CSE
Batch - 60
ID : 20220000000066

Submitted to

Farchana Khandekar Kothari
ID: 2024100000475

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Experiment No: 02

Experiment Name: Simple LAN Design and Configuration
~~with Class A net ID.~~ of Inter-
Network Routing.

Introduction:

- While a Local area network (LAN) connects devices within a single broadcast domain real world enterprise network require communication between multiple distinct network segments. In this lab we designed a topology in Cisco Packet Tracer that bridges two separate networks a Class A network and a Class B network. The primary objective was to configure multiple interfaces on a router to facilitate to inter-network communication and to correctly apply OSI Layer 1 Cabling standards.

Objectives:

- To design and implement a multi-network topology.
- To assign and configure Class A (97.0.0.0) and Class B (130.6.0.0) IP addressing schemes.
- To configure a Layer 3 router to act as the default gateway for multiple distinct subnets.
- To apply the correct physical cabling (cooper straight-through vs cooper crossover).
- To verify end-to-end connectivity across different networks using the Temp Ping utility.

Theoretical Background:

1. Network Classes used:

- Class A network: Class A network (97.0.0.0) are designed for extremely large environments. The first octet defines the network portion, utilizing a default subnet mask of 255.0.0.0.
- Class B network: Class B network (130.60.0.0) are utilized for medium-to-large environments. The first two octets define the network portion utilizing a default subnet mask of 255.255.0.0.

2. Inter Network Routing:

- A switch operates at layer 2 and forwards frames only within the same local network. To allow a device on the 97.0.0.0 network to communicate with a device on the 130.60.0.0 network, a layer 3 device like router is mandatory. The router maintains ~~separate~~ separate interfaces connected to each network and routes packets between them based on its internal routing table.

3. Cabling standards:

- Straight-Through Cable: used to connect dissimilar devices like a PC to a switch or a switch to a router.
- Crossover Cable: used to connect similar ~~to~~ devices directly to one another like a router directly to a server or PC.

4. Network Topology and Hardware Inventory:

The simulated topology consists of two distinct network segments bridged by a central router.

- Hardware utilized: 1x Cisco router, 1x Cisco 2960-24TT switch, 1x PC, 1x Laptop and 2x Servers.
- Transmission media: we used copper straight through for connecting PC, Laptop, servers and router to the central switch ~~and~~ and copper cross-over to connect router directly to server 1.

5. IP Addressing Table:

Based on the network labels the router interfaces act as the default gateways for their respective network segments.

Device Hostname	Interface	Assigned IP Address	Subnet mask
Router0	Fa0/0	97.0.0.1	255.0.0.0
Router0	Fa1/0	130.60.0.1	255.255.0.0
PC0	Fa0	97.0.0.2	255.0.0.0
Laptop0	Fa0	97.0.0.3	255.0.0.0
Server0	Fa0	97.0.0.4	255.0.0.0
Server1	Fa0	130.60.0.2	255.255.0.0

Router Configuration:

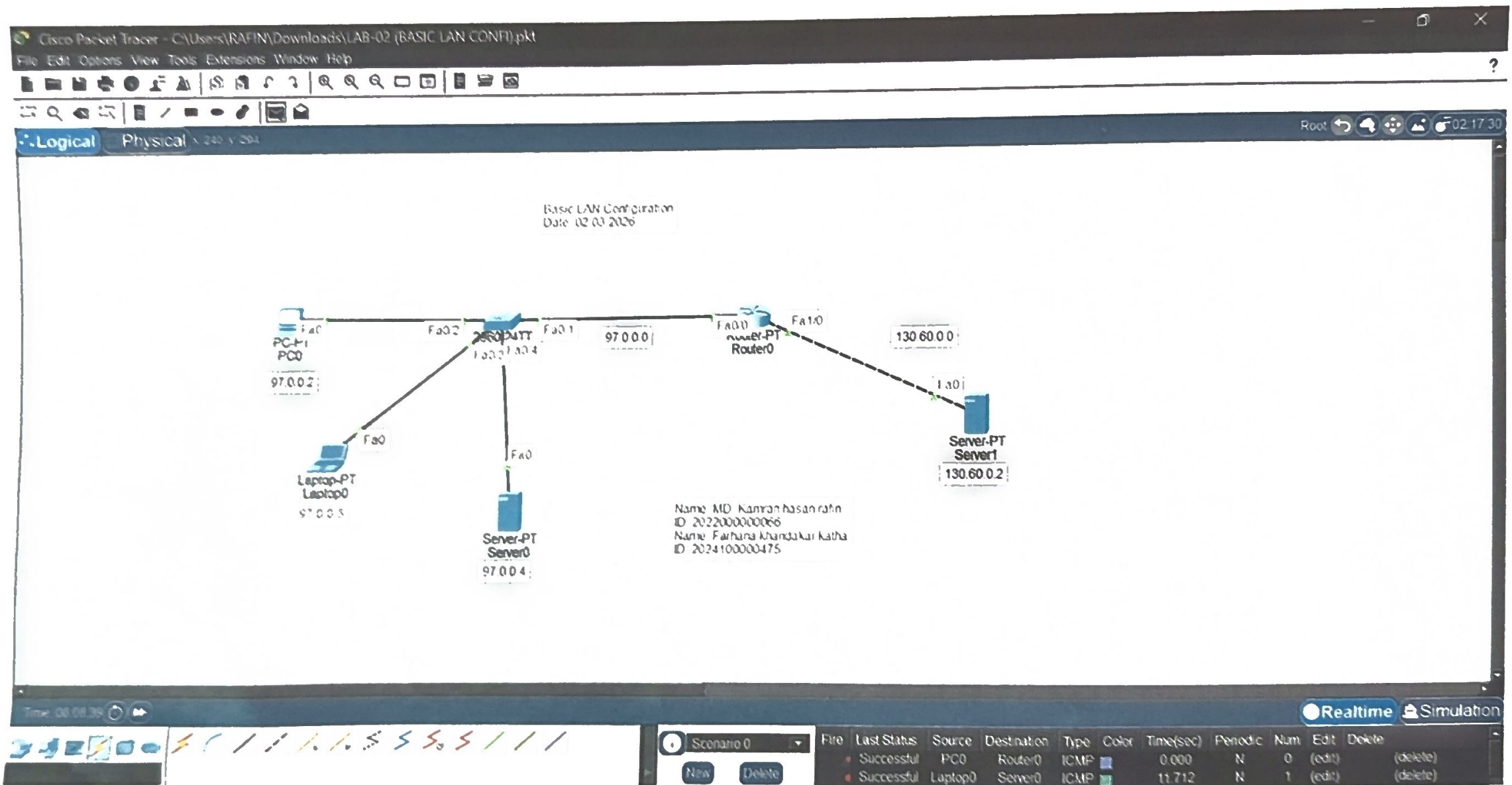
- The router requires IP configurations on two separate Fast Ethernet interfaces to route traffic between the Class A and Class B networks.

1. Access the CLI and enter global configuration mode: enable → configure terminal.

2. Configure Fa0/0 (Class A network):

- Interface fa0/0

- IP address 97.0.0.1 255.0.0.0



- no shutdown.

3. Configure fa 1/0 (Class B network):

- Interface fa 1/0
- IP address 130.60.0.1 255.255.0.0
- no shutdown.

4. End-Device Configuration:

- Open the device configuration window and navigate to desktop → IP configuration.
- Assign the static IP addresses and default Subnet masks according to the addressing table.
- Configure the default gateway on all end devices. This step is critical without it packets destined for the opposite network will be dropped by the originating device.

14] Results and Conclusion:

The multi-network topology was successfully designed and configured. As evidenced by the Packet Tracer Simulation event list, localized and inter-network ICMP connectivity tests were executed successfully.

Through this lab exercise, the core principles of inter-network routing and IP addressing across different classes were successfully demonstrated. The practical application of correct physical cabling was solidified by utilizing a crossover cable for the direct router-to-router connection.

Ultimately the experiment validated how a single routing device can be provisioned to serve as the gateway for multiple distinct subnets.